

A Decision-Theoretic Extension of Causal Unit Selection: Identifiability, Threshold Rules, Strategic Feedback, and the Fallacy of Naive Benefit Rates

Version 4 - strategic extension of the T5-T9 framework

Abstract. The problem of causal unit selection aims to target interventions by maximizing net benefit across unobserved response types. This paper extends the Li-Pearl (2021) benefit function into a decision-theoretic framework. Versions 1-3 establish identifiability, strict dominance thresholds, reliability under estimation, and an autonomy gate for deciding when an AI agent should act versus defer to a human. Version 4 studies what happens after such a rule is deployed into a strategic environment. Once a threshold becomes observable, the quantities used by the rule need not remain exogenous: fraudsters can learn a cutoff, applicants can improve gameable features, and interacting algorithms can adapt. This creates a Lucas/Goodhart feedback problem. We introduce a re-estimation layer over the existing T5-T9 structure and state four strategic extensions (T10-T13): stale deployment can create active error, abstention removes the incentive to manipulate the agent-specific threshold, iterative re-estimation converges within the specified monotone response class, and the equilibrium rule has zero active error. Monte Carlo experiments, two structurally different response families, and public FiFAR and Lending Club data illustrate the mechanism. A key scope result is that strategic feedback is consequential primarily near the decision margin; wide structural margins can make the original gate robust by construction. Uniqueness of the fixed point remains open in the general monotone class.

1. Introduction and Causal Framework

In individualized decision-making, the Average Treatment Effect (ATE) is insufficient for decisions that depend on who benefits and who is harmed. The net outcome is represented by four potential response types: Benefited (B), Always-taker (A), Never-taker (N), and Harm (H).

Li and Pearl (2021) formulate the unit-selection problem using the benefit function:

$$f(c) = \beta \cdot P(B|c) + \gamma \cdot P(A|c) + \theta \cdot P(N|c) + \delta \cdot P(H|c)$$

This structure decomposes as $f(c) = W(c) + \sigma \cdot \text{PNS}(c)$, where $W(c)$ is computable from observable causal-effect margins and PNS is the Probability of Necessity and Sufficiency, $P(B|c)$. The structural master coefficient is:

$$\sigma = \beta - \gamma - \theta + \delta$$

2. Decision-Theoretic Foundations

The first layer of the framework converts partial causal identification into explicit decision rules under structural uncertainty.

Theorem 5 (Identifiability)

Statement. The function $f(c)$ is point-identifiable purely from the causal effects if and only if $\sigma = 0$.

Proof. When $\sigma = 0$, $f(c) = W(c)$, which is a pure effect expression. When $\sigma \neq 0$, a counterfactual shift $\Delta > 0$ between $P(B)$ and $P(H)$ changes the value of the function by $\sigma \cdot \Delta \neq 0$ while leaving the observable effect margins unchanged. QED.

Theorem 6 (Decision Criterion)

Statement. Relying only on observable effects, with $PNS \in [0,1]$ unknown, target groups c_0 and c_1 are strictly decidable if and only if $|W_1 - W_0| > |\sigma|$.

Reason. Strict separation requires the gap between W values to exceed the maximum structural interval length induced by σ . If the intervals overlap, the framework does not license a strict selection and returns ABSTAIN.

Theorem 7 (Reliability Under Estimation)

Statement. Let the true decision margin be $D = |W_1 - W_0| - |\sigma| > 0$. As estimation error shrinks, the probability of an active false selection approaches zero, and the sample requirement scales on the order of $1/D^2$. When $D \leq 0$, more data cannot remove the structural non-identifiability represented by the overlap.

3. LLM Agent Autonomy Guardrail

Version 3 maps the decision criterion to an autonomy gate. Instead of comparing two target groups, the gate compares an agent action with a human-defer baseline while retaining explicit utility weights for benefit and harm.

Theorem 8 (Autonomy Criterion)

Statement. An agent action is strictly and decidable better than deferring to a human baseline if and only if:

$$|\gamma - \delta| \cdot |p_a - p_h| > |\sigma|$$

Here p_a is the agent success rate and p_h is the human success rate. The comparison is independent of the status quo - what would happen if neither acted. Operationally, the rule is: ACT autonomously if the inequality holds; otherwise ABSTAIN and defer to the human.

Theorem 9 (Abstain Safety)

Statement. Under the T8 gate, the agent defers whenever the observable advantage does not dominate the structural uncertainty margin. The explicit harm weight enters through σ and therefore changes the evidence required to unlock autonomous action.

Version 3 validated this gate on the TWINS dataset, public financial-market data, and FiFAR/OpenL2D. The FiFAR experiment reported correct ACT/ABSTAIN decisions in all 54 evaluated segments. The earlier empirical layer also showed how the naive benefit rate $a - b = PNS - P(H)$ can substantially understate utility when benefit and active harm cancel in the observed margin.

4. Version 4: Strategic Feedback After Deployment

4.1 Why the Version 3 parameters can become endogenous

The Version 3 rule is computed from observed margins p_a and p_h . In a passive environment, these can be treated as externally generated estimates. Deployment changes that assumption when the environment can observe or infer the rule. Fraudsters can learn the threshold, credit applicants can improve gameable features, and other algorithms can react to the deployed policy.

This is the relevant connection to the Lucas critique: parameters estimated under one policy need not remain invariant after the policy changes behavior. It is also a Goodhart-type mechanism: once a measured quantity becomes part of the decision criterion, strategic actors can optimize the measured quantity rather than the underlying quality that the quantity was intended to represent.

Important clarification. σ does not change strategically. It is a utility constant. Strategic adaptation changes p_a , p_h , and the effective uncertainty surrounding what those observed margins mean after deployment. Thus the new problem is not a new benefit function; it is an endogenous-data problem

layered on top of T5-T9.

4.2 Stackelberg formulation

- Leader (defender): commits to a threshold rule R .
- Follower (strategic actor): observes R and chooses effort e - evasion, inflation, or another adaptation - to maximize its own payoff. This response determines post-deployment margins $p_a(\tau)$ and $p_h(\tau)$.
- Two directions of response: degradation can reduce apparent agent advantage and close the gate, which is conservative; inflation can raise the apparent p_a without reducing true risk, which is the dangerous direction because stale estimates can keep the gate open.

5. Theorem 10 - Abstain Robustness Under Strategic Adaptation

Statement. Within the specified monotone response class, strategic adaptation can create nonzero active error for a naive Version 3 deployment that freezes pre-deployment parameters. A robust Version 4 deployment that re-estimates the post-response margins to a fixed point retains zero active error in the evaluated configurations.

Derivation. The stale rule evaluates $|\Delta W|$ using margins that were valid before the follower responded. Once p_a and/or p_h become endogenous to the deployed threshold, the stale estimate no longer describes the environment in which the action is taken. This opens a Lucas channel through which an ACT decision can be based on the wrong post-deployment margin. Version 4 closes that channel by repeatedly re-estimating the margins generated by the current rule and updating the rule until the state and the decision criterion are mutually consistent. At that fixed point, the original T8/T9 criterion is evaluated on the margins that actually prevail under the deployed rule.

Monte Carlo verification (v4_validate.py)

Configuration	Naive v3 active errors	Robust v4 active errors
$\sigma = 20$	9 / 20	0
$\sigma = 0$	18 / 20	0
$\sigma = 0$, second setting	17 / 20	0
$\sigma = 20$, second setting	12 / 20	0
$\sigma = 60 / 180$	Gate closed from initialization	0 / structurally protected

The $\sigma = 60/180$ configurations provide a useful negative case: the gate is already closed, so strategic feedback cannot induce an autonomous error. This anticipates the scope result in Section 9.

6. Theorem 11 - Abstention Is Strategically Stable

Statement. If the rule returns ABSTAIN and the strategic effort e only targets the agent-specific threshold, then the follower's best response is $e^* = 0$ and the agent-targeted distribution does not move.

Proof. Let the follower payoff be $U(e) = P(\text{agent decision passes} \mid e) \cdot V - c \cdot e$, with $c > 0$. Under ABSTAIN, the agent does not produce the autonomous outcome. Therefore $P(\text{agent decision passes} \mid e) = 0$ for every e that only manipulates the agent gate. Hence $U(e \mid \text{ABSTAIN}) = -c \cdot e$, which is maximized at $e^* = 0$. QED.

Strategic implication. Abstention is not only a safety state in the T9 sense. Under the stated targeting assumption, it is also incentive-removing: once the autonomous gate is not used, spending

effort to game that gate has no return. This result does not require monotonicity of the response function.

7. Theorem 12 - Convergence of Re-estimation

Statement. Consider the Version 4 update $\tau_{t+1} = F(\tau_t)$. Assume that $p_a(\tau)$ is continuous and non-increasing in τ and that p_h is constant or non-increasing. Within the update class used here, the re-estimation sequence converges to a fixed point τ^* .

Argument. If the gate enters ABSTAIN, Theorem 11 removes the incentive for further agent-targeted manipulation and the induced distribution stops moving, yielding a fixed state. If the gate remains in ACT, the specified monotone response makes the relevant gap evolve monotonically under the update rule; the state is bounded, so the iteration converges to a limit that is self-consistent under F . QED.

Computational verification (v4_validate.py)

The state was implemented as a quantile position q rather than as an ambiguous raw threshold: $F(q)$ is the quantile position in the new distribution corresponding to a fixed risk cutoff. Two structurally different response families - linear and logistic/saturating - were evaluated at three response scales and from five initializations each. All 30/30 runs converged to the same observed $q^* \approx 0.999$, with zero active error.

8. Theorem 13 - Zero Active Error at Equilibrium

Statement. At a Version 4 fixed point τ^* , the rule either returns ABSTAIN or takes the correct autonomous action under the equilibrium margins; therefore $\text{active error}(\tau^*) = 0$.

Argument. At a fixed point, the margins used by the rule are the margins induced by that same deployed rule: $(p_a(\tau^*), p_h(\tau^*))$. If the gate returns ACT, then $|\gamma - \delta| \cdot |p_a - p_h| > |\sigma|$ at the equilibrium state, so the validated T8/T9 decision criterion applies to those equilibrium margins. If the gate returns ABSTAIN, the agent takes no autonomous action and therefore creates no active autonomous error. QED.

9. Validation on Public Data

9.1 FiFAR / OpenL2D - a negative but informative result

The FiFAR configuration has approximately 12% fraud prevalence, so a trivial majority rule is already about 88% accurate. In the evaluated setup, synthetic human analysts at approximately 28-46% accuracy are below that trivial benchmark. This makes it difficult for the agent to become worse than the human baseline through the tested strategic shift. The naive rule therefore remains structurally robust in this regime.

Scope finding: the Goodhart/Lucas failure mode is most relevant near a narrow decision margin. When the baseline imbalance and/or $|\sigma|$ create a wide safety margin, the gate can remain closed or remain safely separated despite strategic adaptation.

9.2 Lending Club - narrow-margin failure and robust re-estimation

The Lending Club experiment uses approximately 1.34 million completed loans with about 20% default. FICO separates observed risk in the sample, with default rates ranging from approximately 26.8% to 11.1% across the evaluated score range. The strategic intervention models feature inflation: a borrower improves gameable variables such as debt-to-income ratio and inquiry count while the underlying true risk is held fixed.

Margin	Naive v3	Robust v4
$\sigma = 0$ (narrow)	9-11 / 12 active errors across three configurations	0

Margin	Naive v3	Robust v4
$\sigma = 20$ (wider)	No failure in the tested bounded-improvement setup	0

The result is deliberately asymmetric: Version 4 is not claimed to make every environment fragile. Instead, the experiment identifies the regime in which stale deployment matters - a narrow enough margin that strategic inflation can cross the autonomy boundary.

9.3 Stackelberg verification (v4_validate.py)

- Observed response monotonicity in the experiment: p_a decreases from 0.717 to 0.237 as the strategic threshold state increases.
- Five initializations converge to $\tau^* \approx 0.116$ in 5-7 update steps.
- The observed fixed point remains in ACT: $p_a = 0.704$ and $p_h = 0.317$, with zero active error under the equilibrium evaluation.
- A placebo response is used as a negative control and does not converge to the same fixed point.

10. Open Boundary: Existence and Convergence Do Not Establish Uniqueness

The Version 4 result is intentionally limited. T12 supplies a convergence result for the specified update class and the experiments repeatedly reach a fixed point, but this does not prove that the fixed point is unique over the general monotone response class.

The tested update maps are not contractions. The measured Lipschitz constants range from 1.10 to 1.74, all above 1. Therefore Banach's fixed-point theorem cannot be used to establish uniqueness in the general class considered here. Empirically, all tested initializations in the two response families converged to the same point (30/30), but formal uniqueness remains an open mathematical boundary rather than a claimed theorem.

11. Provenance, Scope, and Limitations

- Relationship to prior versions. T10-T13 are strategic corollaries and deployment extensions of T5-T9. Version 4 does not replace the original unit-selection mathematics; it adds an endogenous-response layer and tests whether the original decision gate remains valid after deployment.
- Monotonicity. T12 is restricted to the stated monotone response/update class. A non-monotone or "grazing" adversary lies outside this phase of the analysis. T11 remains applicable without monotonicity as long as effort only targets the autonomous agent gate.
- Human baseline. Calibration of p_h in the simulations is part of the test design used to place the naive system initially in ACT. It is not a claim about real human performance in the application domain.
- Lending Club model. The strategic simulation is a simplification using a linear-logit risk model and a fixed improvement mechanism. It demonstrates the feedback channel; it is not a structural model of borrower behavior.
- Reproducibility package. Version 4 retains the data sources and supporting materials already archived with Version 3. The only new code file added for Version 4 is v4_validate.py, which contains the Version 4 Monte Carlo, response-family, public-data, and Stackelberg validation routines reported in this paper.

12. Conclusion

Version 4 extends a static autonomy guardrail into a strategic deployment setting. The central result is not a new benefit function. It is a deployment principle: a threshold derived from causal decision margins can become invalid when the threshold changes the data-generating process. Freezing pre-deployment margins can therefore create active error near the boundary. Re-estimating the induced margins to a self-consistent fixed point restores the original T8/T9 decision logic in the evaluated monotone settings.

Abstention plays a second role in this extension. Beyond protecting against structural uncertainty, it can remove the payoff from manipulating an agent-specific threshold. This makes ABSTAIN both a decision-theoretic safety state and, under the stated targeting assumption, a strategically stable state. The empirical boundary is equally important: strategic feedback is not uniformly dangerous. Wide margins can make the gate robust by construction, while narrow margins are the regime in which Goodhart/Lucas feedback must be modeled explicitly.

The remaining theoretical boundary is uniqueness. The observed update maps are not contractions, so the repeated convergence to a common fixed point in the tested families does not establish a unique equilibrium for the full monotone class. That question is left open rather than assumed away.

Version 4 change summary

- Retains the Version 3 causal unit-selection framework and T5-T9 autonomy gate.
- Adds strategic endogeneity of p_a and p_h after deployment; σ remains a fixed utility constant.
- Adds T10-T13: stale-parameter failure, strategic stability of abstention, fixed-point re-estimation, and equilibrium zero-active-error.
- Adds Monte Carlo, two-family response verification, Lending Club validation, Stackelberg checks, and a negative FiFAR scope result.
- States the non-uniqueness boundary explicitly: measured Lipschitz constants exceed 1, so Banach contraction arguments do not apply.

References and Data Sources

Li, A. and Pearl, J. (2021). Causal unit selection / benefit-function framework. The present paper uses the benefit-function decomposition and PNS terminology from that work.

Version 3 data sources and supporting materials are retained as previously archived. Version 4 adds one new validation file only: v4_validate.py. This single script contains the Version 4 validation routines referenced throughout Sections 5-10.